

The Classical and Quantum Dynamical Simulations

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Solving quantum systems for varied purposes, especially in experimental structures and understanding of quantum system in theory or in practical application. This study emphasizes and focus on various simulation methodology, theoretical foundations and development of the same field of operation, practical problems and general overview of the performance, usefulness, and overall meaning of simulation in quantum mechanics.

1 Outlining

The focal of this section is on the topic itself at large. Of such, the topic I propose here would be of **quantum mechanics**, specifically its theory and formulation, as well as certain specific baseline case, such as **optical spectroscopy** setting, or rather, expository into quantum optics. However, to keep things simple - it would be mostly of quantum physics.

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2 Motivation

Because the outlining section is non-standard, and the whole document is in the spirit of a draft, thereby, the full detail would likely to be explicitly included.

2.1 The Classical Mechanical Formalism

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2.2 The Quantum Mechanical Formalism

The quantum mechanical formalism, as well as the general quantum theory, is settled within experiments, of which comes the **axioms** and **postulates** of assumptions to the quantum system (such as encoding the state of a quantum system, in a operating space of parameter x , and given time t , is the wavefunction $\Psi(x, t)$); the **definition** of the object in such quantum system, the **spaces** of dimensions within which quantum physics is enabled, or rather specified as the proxy of which the physically realizable space is defined, the evolutionary rules of which is encoded by the Schrodinger equation and, to certain rejection of such equation, the Born rule for the interpretation of the ψ -function.

2.2.1 Axioms of quantume mechanics

As of such, we can list a few "postulate" of formalism:

- I The state of a quantum mechanical system, including all the information you can know about it, is represented mathematically by a normalized ket $|\psi\rangle$
- II To every observable in classical mechanics there corresponds a linear, Hermitian operator in quantum mechanics.
- III In any measurement of the observable associated with operator $\hat{A}$, the only values that will **ever** be observed are the eigenvalues, a that satisfy the eigenvalue equation $\hat{A}\Psi = a\Psi$.
- IV If a system is in a state described by the normalized wavefunction (as to be **physically realizable**) Ψ , then the average value of the observale corresponding to $\hat{A}$ is given as

$$\langle A \rangle = \int_{-\infty}^{+\infty} \Psi^* \hat{A} \Psi d\tau$$

- V The wavefunction or state function of a system evolves in time according to the time-dependent Schrodinger equation.
- VI The total wavefunction must be **asymmetric** with respect to the interchange of all coordinates of one fermion with those of another. Electron **spin** must be included in this set of coodinates.

As one might see, those are **conditionals**, as well as differential evolution (the "law" of the evolution of state). Of such, the confinement was reached. Can we figure out in some way or another, to randomly simulate such processes, and said system?

2.2.2 The operational observations

By the nature of the concept and formalism in its history, quantum mechanics works as a proxy of the physically realizable world, through the lenses of several mathematical formalism:

1. Linear algebra - the proxy **spaces** and representation, as well as the behaviour of **superpositions** of the object, as well as its presenting solution of the system.
2. 2. Probability theory - the uncertainty of measurement, observations, and limit to accuracy estimation (the law of the middle element).
3. Operator theory - the characteristic inner transformation of the proxy space.
4. Clifford algebra - the necessity for such algebraic manipulation (kind of) within vector spaces and its dual spaces of state transition.
5. Analysis and analytical form on Hilbert space (of infinite continuous differentiable space) - The analysis of infinitesimal scale and the assumption that every "position" in the quantum encoding space is **extractible** of continuous nature.
6. Differential equation, specifically, the standard Schrodinger equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

presenting the *implicit representation* of the dynamical changes and evolution of the system. (What can be such implicit representation?)

7. Potential representation $V(x)$ taken from Hamiltonian formalism instead of direct force factor in Newtonian classical mechanics.
8. Complex analysis - giving out the **interpretation** of the wavefunction $\Psi(x, t)$.

Following the above notions, we note the following point of such observation:

1. In a sense, we can inspect the physical world alone, from the quantum mechanical proxy, albeit only certain specification would make it physically realizable.
2. Not everything in quantum mechanical formulation is **definite** and in closed-form solution, rather the description of such state relies on postulates and axioms, of which the implicit representation and process is rather refined.
3. Similar (to certain point) as (2.), most of the quantum mechanical solution is not in closed-form solution, but the solution, at least to the evolutionary state of Schrodinger equation, is categorized by numerical analysis and iterative solutions.
4. The nature of the time-evolution system, is directly anecdotal to the **stochastic** process.

Thereby, it is consequential, and rather, logical, to assume certain fixture toward mathematical modelling process of the quantum system, and specially, **quantum simulation** of the quantum state (Hangleiter et al. [2022]).

3 The clarification of simulation

The term “quantum simulation” is used variedly in academic and industry. For most of its references, of Bauer et al. [2022], Buluta and Nori [2009], Fraxanet et al. [2022], Georgescu et al. [2013], Hangleiter et al. [2022], Sanchez-Palencia [2018], it is referred to as the “quantum simulation of another quantum system”. In rather some cases, it is referred to as rather a numerical solution, such that the quantum Hamiltonian is realized, but in a solicited, discrete form - for example, in **quantum chemistry**, it is the treatment of the molecular Hamiltonian of atoms model - then using the algorithm to calculate either its finite stationary state, or a low and local amount of time-evolution of such system.

Here, we want to clarify our purposes regarding such quantum simulation system.

1. By quantum simulation we means two types of those: For once, we mean the quantum dynamical simulation, of which simulates the dynamics of a physical system under quantum mechanical law. This is via the manipulation of Hamiltonian term for each object, Schrodinger equation and von-Neumann equation (for mixed states) to govern the infinitesimal time-evolution jump state of the quantum system. This system only uses the quantum mechanical law established, within its axioms, and thereby are not subjected substantially to experimental verifications. In fact, they are **theoretical models**, of which like the case of a classical physics engine, we are actually doing the quantum physical engine, purposefully. Classically, we have
 - i Quantum Molecular Hamiltonian simulation of the molecular state (particle theory)
 - ii Specific Hamiltonian operator and numerical solution emulation (Ising model, Monte Carlo method, statistical inferences).

all of which can be done with classical computers. Controlling the behaviours and else would use **machine learning** or other “concept learning” method to process the data.

2. The second is the more anecdotal and empirical mean, the quantum simulation of different simplified system, of such that it is a simplified form of certain specialized cases and environment, with realizable Hamiltonian and system mechanics. More precisely, this does not only contain dynamical simulation (as we have seen above), but more generalized:

More precisely, a quantum simulator is a controlled device that allows us to (i) engineer a class of quantum Hamiltonians exactly, (ii) control its dynamics, and (iii) make sufficiently precise measurements to consider that the problem is solved. Sanchez-Palencia [2018]

This method requires the feedback loop of verifying to the environment itself - that is, requires experiments to be conducted within the same ensemble as the model (in fact, it is quite the reverse in most cases) that would be then feed to the simulated model itself, so that there exists certain supervisor to change the configuration of the simulated system.

3. Another method is to examine medium and light quantum system, with certain fixed size and theoretical circumstances (for example, the infinite square well), using the proposed method as the **agent injection** method - simply by adding test agent into an volatile quantum system, but generally stable in accuracy of behaviours. This is analogous to trying to test on the three body problem using the reduced two-body problem setup, for analogy.

The first one is a purely theoretical model, rely only on computation and ideal mass-test of ideal quantum system, based of the axioms, postulates and other “behavioural controller” of the evolving system. The second one is a more feedback loop, continue

On the side note of such I also want to have a different way of looking into experiments of quantum mechanical system, as far as textbooks and articles are concerned of (old paper, that is). One of such is the treatment of *phase spaces* and *layered abstraction* of quantum mechanical treatment upon the formalism into operator theory - operators that acts on certain system. The presumption of *layered abstraction* is because quantum mechanical description is also an abstraction onto the physical raelizable world: for example, we inherently agreed that $|\Psi(x, t)|^2$ is the probability that the wavefuction represent the occurence of such particle, and the condition for it to be **physically realizable solution** is such that

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 0 \quad (1)$$

which in itself is exactly the upper layers abstraction to the philosophy of *absolute determinism* of classical formalism - which unfortunately failed.

4 Provisional Goal

The goal of the current design theory is to formalize (and study) the theoretical mechanical philosophy and its formality, then to the quantum (modern) theoretical formality of interest. The comparison between those two, in term of all postulates and propositions, assumptions and observations, should be realized, in almost everything that has been mentioned or related, within the scale of mechanics and abstract kinematic (perhaps only classical theory can have the *virtue* of using kinematic without actions).

The most likely actual concern would be the **computational powers** needed to simulate such system. Overall, it is hard to say but the nature of quantum mechanics makes it so that everything seems to be very complicated and complex, so much so that the computational power is not usually enough for a large sample test.

5 First Methodology Draft September 2024

5.1 Classical Dynamical Simulations

This is most likely in the form of a **theoretical physic engines**, thoses with same structures as the normal classical simulation and game theory engines.

5.2 Quantum Dynamical Simulation

For quantum dynamical simulation, the first idea comes to mind, as a straightforward, unverified line of thoughts contains several key points:

I Visualization and structural representation

- (a) The structural and empirical analysis of the process.
- (b) The simulating techniques: we have to cases to be considerate of: **multi-agent** or **single-agent**.
- (c) In how many ways can be visualize quantum mechanical system?

II *Simulating basis* Creating the code base for simulation, and test out performances and metrics.

III Theoretical framework

- (a) Setting up the postulates and axioms. Definitions and transition rules, components of the quantum system of simulation.
- (b) Creating, and theorizing various combination of such postulates and axioms to justify the theory of quantum mechanics.
- (c) Probabilistic assumption, of which the description of quantum mechanics is settled - elementarily, it's $|\Psi^2|$ and things as such.
- (d) Monitoring parameters: the momentum, position, potential, etc.

IV Simulation process layout

- (a) Setting up, and writing down the supposed and theoretical process of running simulation. Certain criteria like time-dependent agent updating procedural.

The goal of these works are to set up the required theoretical background as well as theoretical framework of interest. These both task are used for both learning and making - learning quantum mechanics, and making research on quantum mechanics too.

The codebase also must be complete - this means the code and general experiment setting that will support various experimental process later on. For example, for the general Schrodinger equation:

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

We notice that there is the implicit component $\Psi(x, t)$, the potential $V(x)$ as a moving part, and relational condition as all the remained terms. Then

- i We will have one module $pot(x)$ for inputting various potentials into play - this can be represented as the potential landscape of the usual space. Plus, usually, the potential is in 2-dimensional, hence, a contour convolutional upon the normal space should be fine.
- ii A representative module $wfunc(x, t)$ of the wavefunction.
- iii A total module $config(potential, wfunc)$ taking both term as argument.
- iv A $t + 1$ time-increment module $evolve(config)$ of which we take on the full form of the Schrodinger equation (TDSE).

Note that usually, $wfunc(x, t)$ is not explicitly defined, so implicit definition should be valid (in accordance to the general solution). Furthermore, we might be able to use a technique as **restricted-well release** - predefining the wavefunction in a ground mode - for example, of the infinite/finite square well $V_{\perp}(x)_A$, then release the barrier and expect certain behaviour to the more complex, of-target potential well $V'(x)_D$, for A and D being the "boxed interval", such that $|A| < |D|$ as for the scale of such. Angelakis [2017]

Moving on, we would want to focus heavily on actual simulating tests and thereof.

- 1. The old way - closed-form solution, combination of possible solution of the space, usually speaking, it is that you just derive everything, and solve it analytically.
- 2. The numerical way - instead, going for approximated solution - and iteratively get closer to the acceptable results.
- 3. The "new" (at least I'm not aware of) way: Random simulation techniques and settings.

For that, we would like to then choose possible configuration of the quantum system, according to certain setting, like optical system with its specific properties, or so.

Certainly, we would also want to simulate the optical process - for example, the Raman scattering process. How can we do such, at least in term of the codebase given, and how would one simulate it effectively of a en masse collision event?

This gives us quite a few options, but also quite a few problems. However, it is interesting to look at it - by simulating also, depends on interpretations - for example, the **vibrational frequency** explanation, or the **virtual state** and energy absorption explanation would be an interesting touch.

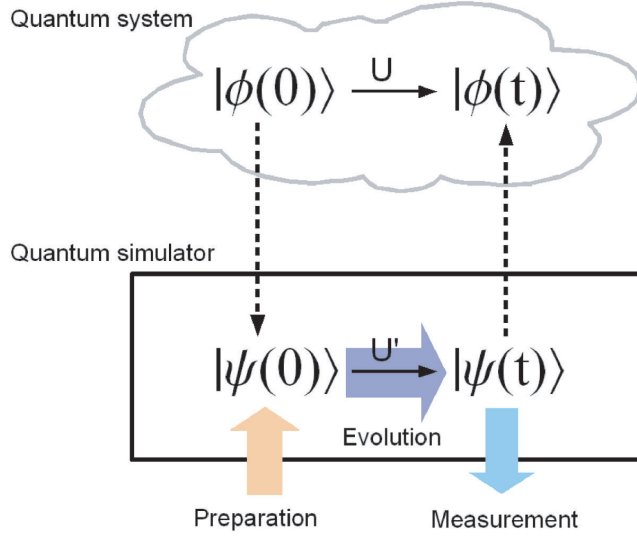


Figure 5.1: (color online) Schematic representation of a quantum system and a corresponding quantum simulator. The quantum state $|\phi(0)\rangle$ evolves to $|\phi(t)\rangle$ via the unitary transformation $U = \exp\{-i\hbar H_{\text{sys}}t\}$. The quantum simulator evolves from the state $|\psi(0)\rangle$ to $|\psi(t)\rangle$ via $U' = \exp\{-i\hbar H_{\text{sim}}t\}$. The simulator is designed such that there is a mapping between the simulator and the simulated system, in particular the mappings $|\phi(0)\rangle \leftrightarrow |\psi(0)\rangle$, $|\phi(t)\rangle \leftrightarrow |\psi(t)\rangle$ and $U \leftrightarrow U'$. While the simulated system may not be controllable (or not experimentally accessible in some cases), the quantum simulator is. Namely, the initial state $|\psi(0)\rangle$ can be prepared, the unitary evolution U' can be engineered, and the final state $|\psi(t)\rangle$ can be measured. The result of this measurement provides information about the simulated system. The color arrows denote the controllable operations. The solid black arrows describe the time evolution of the system and the simulator. The dashed arrows indicate the correspondence between the quantum states of the simulator and the simulated system. Courtesy of Georgescu et al. [2013]

Under the direct definition of quantum simulations as a two-way process by Georgescu et al. [2013], and per Buluta and Nori [2009]: *By quantum simulator, we understand a controllable quantum system used to simulate emulate other quantum systems*, it is imperative that we need to couple our simulated system with an actual quantum mechanical system. If we based of our theoretical and propositional stance toward Feymann's idea of the quantum simulated machine, simulating quantum system by quantum mechanical means, then there is certainly a need for an accessible quantum system as to prevent the case of inaccessible experiment as far as possible, if there is such chance for quantum experiments of available resources.

Of such, the option, or rather the choice of simulating a scattering events, or several low-complexity to medium-complexity, medium stable quantum system, and those that can be simulated using minimal material - for example, the AlGaAs semiconducting crystal, for example, as in the case of a finite semiconducting barrier cell. One of such option, as well as considering the historical application of classical quantum simulation (quantum simulation on classical computer and computing algorithm), Raman spectroscopy, or others spectroscopy type within variational quantum scale effects, is enough for direct experiments and stable controls.

References

- Ehud Altman, Kenneth R. Brown, Giuseppe Carleo, Lincoln D. Carr, Eugene Demler, Cheng Chin, Brian DeMarco, Sophia E. Economou, Mark A. Eriksson, Kai-Mei C. Fu, Markus Greiner, Kaden R. A. Hazzard, Randall G. Hulet, Alicia J. Kollar, Benjamin L. Lev, Mikhail D. Lukin, Ruichao Ma, Xiao Mi, Shashank Misra, Christopher Monroe, Kater Murch, Zaira Nazario, Kang-Kuen Ni, Andrew C. Potter, Pedram Roushan, Mark Saffman, Monika Schleier-Smith, Irfan Siddiqi, Raymond Simmonds, Meenakshi Singh, I. B. Spielman, Kristan Temme, David S. Weiss, Jelena Vuckovic, Vladan Vuletic, Jun Ye, and Martin Zwierlein. Quantum Simulators: Architectures and Opportunities, December 2019. URL <https://arxiv.org/abs/1912.06938v2>.
- Dimitris G. Angelakis, editor. *Quantum Simulations with Photons and Polaritons*. Quantum Science and Technology. Springer International Publishing, Cham, 2017. ISBN 978-3-319-52023-0 978-3-319-52025-4. doi: 10.1007/978-3-319-52025-4. URL <http://link.springer.com/10.1007/978-3-319-52025-4>.
- Christian W. Bauer, Zohreh Davoudi, A. Baha Balantekin, Tanmoy Bhattacharya, Marcela Carena, Wibe A. de Jong, Patrick Draper, Aida El-Khadra, Nate Gemelke, Masanori Hanada, Dmitri Kharzeev, Henry Lamm, Ying-Ying Li, Junyu Liu, Mikhail Lukin, Yannick Meurice, Christopher Monroe, Benjamin Nachman, Guido Pagano, John Preskill, Enrico Rinaldi, Alessandro Roggero, David I. Santiago, Martin J.

- Savage, Irfan Siddiqi, George Siopsis, David Van Zanten, Nathan Wiebe, Yukari Yamauchi, Kübra Yeter-Aydeniz, and Silvia Zorzetti. Quantum Simulation for High Energy Physics, April 2022. URL <https://arxiv.org/abs/2204.03381v1>.
- Iulia Buluta and Franco Nori. Quantum Simulators. *Science*, 326(5949):108–111, October 2009. ISSN 0036-8075, 1095-9203. doi: 10.1126/science.1177838. URL <https://www.science.org/doi/10.1126/science.1177838>.
- Joana Fraxanet, Tymoteusz Salamon, and Maciej Lewenstein. The Coming Decades of Quantum Simulation, April 2022. URL <https://arxiv.org/abs/2204.08905v2>.
- I. M. Georgescu, S. Ashhab, and Franco Nori. Quantum Simulation, August 2013. URL <https://arxiv.org/abs/1308.6253v3>.
- Dominik Hangleiter, Jacques Carolan, and Karim P. Y. Thébault. *Analogue Quantum Simulation: A New Instrument for Scientific Understanding*. 2022. doi: 10.1007/978-3-030-87216-8. URL <http://arxiv.org/abs/2303.00814>. arXiv:2303.00814 [physics, physics:quant-ph].
- Laurent Sanchez-Palencia. Quantum simulation: From basic principles to applications, November 2018. URL <https://arxiv.org/abs/1812.01110v2>.